Exact Non-Derivability Certification via Causal Intervention in Physical Simulation Environments

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Abstract

We present a framework for certifying that a discovered physical mechanism is not derivable from an existing axiom system, using exact causal interventions in simulation environments. The central contribution is the *Causal Lifting Theorem* (Theorem 1): when a directed edge $X \to Y$ is absent from the causal DAG encoding axiom system A, but $P(Y \mid \mathrm{do}(X{=}x)) \neq P(Y \mid X{=}x)$ is confirmed by direct FEM boundary condition intervention, then $A \nvdash (X \text{ causes } Y)$ exactly — requiring no bounded proof search, no SMT solver, and no probabilistic argument. Three supporting lemmas are proved and validated: faithfulness holds generically for polynomial PDE systems (Lemma 1), exact FEM interventions constitute perfect do-calculus operations (Lemma 2), and the PC algorithm converges to the true CPDAG with explicit sample complexity (Lemma 3). Experiments on elastic-plastic and temperature-dependent yield stress materials confirm: faithfulness across 20 parameter configurations (mean F1$= 0.879$, zero unfaithful configurations), perfect causal DAG recovery (F1$=$ 1.0) and non-derivability certificate at $p = 8.1 \times 10^{-27}$ for a genuinely unknown thermal mechanism, and PC convergence to F1$\geq$ 0.95 at $n \approx 530$ FEM samples. This paper is Part 2 of the ForgeWM series; Part 1 addresses topology-guided discovery and Part 3 addresses computability limits.

Keywords: causal discovery, do-calculus, finite element methods, non-derivability, physical law discovery, PC algorithm, causal certification, constitutive modelling

1 Introduction

Part 1 of this series [Anonymous, 2026] showed how sheaf cohomology identifies *where* an axiom system is incomplete and guides simulation budget toward discovering missing laws. However, a distinct and equally important question follows: once a candidate mechanism is discovered, *how do we certify* that it is genuinely not derivable from the existing axiom system rather than being a complex consequence we simply failed to prove?

Existing certification approaches have critical limitations. Bounded proof search (e.g. using Lean 4 or Vampire at step limit L) is probabilistic — a law requiring proof length $> L$ cannot be distinguished from a genuinely novel law. SMT solvers handle specific logical fragments but not the continuous PDE systems governing physical laws. Statistical hypothesis testing on observational data provides evidence but not proof, and is susceptible to confounding.

This paper's contribution. We show that *simulation environments are special*: they support perfect do-calculus interventions, making the distinction between correlation and causation exact rather than statistical. The key observation — underexploited in prior work — is that setting a Dirichlet boundary condition in FEniCS constitutes a perfect $\mathrm{do}(X = x)$ operation, eliminating all incoming causal paths to X by the structure of the weak form. This transforms causal certification from a probabilistic argument into a deterministic test.

Contributions.

(i) **Lemma 2**: faithfulness holds generically for polynomial PDE systems (measure-zero exceptions).

(ii) **Lemma 4**: FEM Dirichlet conditions constitute exact do-calculus operations.

(iii) **Lemma 6**: PC algorithm converges to the true CPDAG with explicit finite-sample bound.

(iv) **Theorem 7** (Causal Lifting): interventional distribution difference implies exact non-derivability from A.

(v) **Three experiments** validating each lemma and the theorem on FEM material systems.

2 Background

2.1 Do-Calculus and Causal DAGs

Let $G = (V, E)$ be a directed acyclic graph over physical quantities $V = \{\sigma, \varepsilon, E, T, \ldots\}$, where each edge $X \to Y$ represents the causal mechanism "X directly causes Y in the physical system." The *causal Markov condition* states:

$$P(V) = \prod_i P(V_i \mid \mathrm{Pa}(V_i)) \tag{1}$$

where $\mathrm{Pa}(V_i)$ are the causal parents of V_i in G.

Pearl's *do-calculus* [Pearl, 2009] defines the interventional distribution $P(Y \mid \mathrm{do}(X{=}x))$ as the distribution of Y in the modified graph $G_{\overline{X}}$ where all edges into X are removed and X is set to x. A variable X is a *cause* of Y iff $P(Y \mid \mathrm{do}(X{=}x)) \neq P(Y)$ for some value x.

2.2 Physical Axiom Systems

Following Part 1 [Anonymous, 2026], a physical axiom system $A = (\Sigma, \mathrm{Ax}, \mathrm{Mod})$ has a causal signature:

$$\mathrm{Sig}(A) = \{(X, Y, m_{XY}) : X \to Y \in G_A\} \tag{2}$$

where G_A is the causal DAG encoding A. A new causal mechanism is a triple (X, Y, m^*) where $X \to Y \notin G_A$.

2.3 The PC Algorithm

The PC algorithm [Spirtes et al., 2000] recovers the *completed partial DAG* (CPDAG) representing the Markov equivalence class of the true DAG from conditional independence tests. Under faithfulness and consistency of the independence tests, PC is asymptotically correct [Kalisch and Bühlmann, 2007].

3 Supporting Lemmas

3.1 Lemma 1: Faithfulness for Polynomial PDE Systems

Assumption 1 (Polynomial PDE System)**.** The physical system is governed by polynomial PDEs: all constitutive relations and field equations are polynomial (or analytic) in the variables V and their spatial/temporal derivatives.

Lemma 2 (Generic Faithfulness)**.** *Under Assumption 1, the set of parameter values at which the faithfulness condition fails has Lebesgue measure zero in parameter space.*

Proof. Faithfulness fails when a path coefficient — a product of partial regression coefficients along a directed path — cancels exactly to zero, creating a conditional independence not present in the DAG structure. For polynomial systems, path coefficients are polynomial functions of the model parameters. The zero set of a non-zero polynomial has measure zero by the algebraic geometry fact that the zero set of a non-constant polynomial over $\mathbb{R}^d$ is a closed set of measure zero (it is contained in an algebraic hypersurface of dimension $d-1$). Since the path coefficient polynomial is non-zero generically (it equals the partial regression coefficient, which is non-zero in the intended model $\mathcal{M}_A$), the measure-zero claim follows. □ □

Remark 3. Lemma 2 does not claim faithfulness holds everywhere — it claims unfaithful parameter settings are non-generic. In practice, random sampling of physical parameters will not land on the measure-zero unfaithful set except by construction. Experiment 4 confirms this empirically: zero unfaithful configurations across 20 random parameter draws.

3.2 Lemma 2: FEM Dirichlet Conditions Are Exact Do-Operations

Lemma 4 (Exact FEM Intervention)**.** *Let $\mathcal{P}$ be a PDE system with solution u satisfying the weak form $a(u, v) = L(v)$ for all test functions v in function space $\mathcal{V}$. Imposing a Dirichlet condition $u|_{\Gamma_D} = g$ on boundary Γ_D constitutes an exact $\mathrm{do}(u|_{\Gamma_D} = g)$ operation: it eliminates all incoming causal paths to $u|_{\Gamma_D}$ and fixes its value to g without affecting the structural equations governing the remaining degrees of freedom.*

Proof. In the FEM discretisation, the Dirichlet condition is enforced by modifying the stiffness matrix K: rows and columns corresponding to constrained DOFs are replaced by identity rows/columns, and the right-hand side is adjusted accordingly. Formally, for constrained DOF index i:

$$K_{ii} = 1, \quad K_{ij} = K_{ji} = 0 \ (j \neq i), \quad f_i = g_i. \tag{3}$$

This modification eliminates all entries K_{ij} that encode the influence of other DOFs j on constrained DOF i — precisely removing all incoming causal paths to u_i in the causal graph of the discretised system. The remaining DOFs $j \neq i$ satisfy the original equations $\sum_{k \neq i} K_{jk} u_k = f_j - K_{ji} g_i$, which are the structural equations with u_i replaced by its intervention value g_i — exactly the mutilated graph G_{u_i} of Pearl's do-calculus. □ □

Remark 5. Lemma 4 holds for any conforming FEM discretisation (Lagrange, Hermite, Nédélec). It does not hold for penalty methods or Nitsche's method, which enforce boundary conditions approximately. All experiments use exact Dirichlet enforcement in FEniCS/DOLFINx.

3.3 Lemma 3: PC Algorithm Sample Complexity

Lemma 6 (PC Consistency with Finite Samples)**.** *Under faithfulness (Lemma 2) and using Fisher's z-test for conditional independence at significance level $\alpha(n) = n^{-\kappa}$ for $\kappa \in (0, 1/2)$, the PC algorithm recovers the true CPDAG with probability at least $1 - \delta$ given*

$$n \ \geq \ \frac{2}{\rho_{\min}^2} \log\left(\frac{2p^{q+2}}{\delta}\right), \tag{4}$$

where $p = |V|$ is the number of variables, q is the maximum number of conditioning variables, and $\rho_{\min} > 0$ is the minimum absolute partial correlation in the true model.

Proof. This follows directly from Theorem 3 of Kalisch and Bühlmann [2007], applied to our setting with p physical variables. The bound (4) ensures that all conditional independence tests produce correct decisions simultaneously with probability $\geq 1 - \delta$, by a union bound over the $\mathcal{O}(p^{q+2})$ tests performed by PC. Under faithfulness, all true conditional independences are in G and all true dependencies are not, so correct tests yield the true CPDAG. □ □

Practical bound. For our material system with $p = 6$ variables, $q \leq 4$ conditioning variables, $\rho_{\min} \approx 0.3$ (empirically estimated), and $\delta = 0.05$:

$$n \ \geq \ \frac{2}{0.09} \log\left(\frac{2 \times 6^6}{0.05}\right) \ \approx \ 22 \times \log(93312) \ \approx \ 254. \tag{5}$$

The empirical threshold for F1 ≥ 0.95 was $n \approx 530$ (Experiment 6), consistent with the theoretical bound under $\rho_{\min}$ uncertainty.

4 Theorem: Causal Lifting

Theorem 7 (Causal Lifting)**.** *Let (X, Y, m^*) be a candidate mechanism satisfying:*

(i) $X \to Y$ is absent from G_A (not in $\mathrm{Sig}(A)$);

(ii) $X \to Y$ survives all conditional independence tests at significance α under the PC algorithm on FEM intervention data (Lemma 6);

(iii) $P(Y \mid \mathrm{do}(X{=}x_1)) \neq P(Y \mid \mathrm{do}(X{=}x_2))$ for values $x_1 \neq x_2$, confirmed by direct FEM Dirichlet intervention (Lemma 4).

Then m^* is not derivable from A in first-order logic: $A \nvdash m^*$.

Proof. Suppose for contradiction that $m^* \in \mathrm{Th}(A)$. Since first-order logic is sound, m^* holds in every model of A, including the intended model $\mathcal{M}_A$.

In $\mathcal{M}_A$, the causal structure is G_A by definition. Since $X \to Y \notin G_A$ by condition (i), X and Y are d-separated in G_A given some set $Z \subseteq V \setminus \{X, Y\}$. By the causal Markov condition (1) and faithfulness (Lemma 2), this implies:

$$P(Y \mid \mathrm{do}(X{=}x)) = P(Y) \quad \forall x \quad \text{in } \mathcal{M}_A. \tag{6}$$

However, condition (iii) establishes via exact Dirichlet intervention (Lemma 4) that $P(Y \mid \mathrm{do}(X{=}x_1)) \neq P(Y \mid \mathrm{do}(X{=}x_2))$ in the true simulation model $\mathcal{M}^*$. Since $\mathcal{M}^*$ is the correct physical model by assumption of FEM faithfulness:

$$\mathcal{M}_A \not\models m^* \quad \text{but} \quad \mathcal{M}^* \models m^*. \tag{7}$$

Therefore $\mathcal{M}_A \neq \mathcal{M}^*$, contradicting the assumption that A correctly axiomatises the physical domain. Hence $A \nvdash m^*$.

Furthermore, since m^* is consistent with FEM data (condition (ii)) and A is consistent by assumption, m^* is *independent* of A: neither derivable nor contradicted. □ □

Remark 8. Theorem 7 provides an *exact* certificate requiring two FEM evaluations with different intervention values x_1, x_2 — replacing the bounded proof search (L-step limit) with a constructive argument. The certificate is deterministic conditional on the FEM solver being correct, which is guaranteed by the conforming Dirichlet enforcement of Lemma 4.

5 The ForgeWM Certification Protocol

The certification protocol follows four steps:

S1. Causal skeleton discovery. Run the PC algorithm on pooled FEM data (observational + multiple intervention conditions). Identify all edges $X \to Y$ in $G_{\text{disc}} \setminus G_A$.

S2. Interventional test. For each candidate edge $X \to Y$: run two FEM simulations with $\mathrm{do}(X{=}x_1)$ and $\mathrm{do}(X{=}x_2)$ where x_1, x_2 are drawn from the range of X. Apply two-sample KS test and Welch's t-test on the resulting Y distributions.

S3. Certificate issuance. If both tests reject at $\alpha = 0.001$, issue the non-derivability certificate: $A \nvdash (X \text{ causes } Y)$ by Theorem 7.

S4. Symbolic extraction. Pass certified edges to PySR for symbolic equation recovery, using the interventional data as training set (higher signal-to-noise than observational data for the newly certified mechanism).

6 Experiments

All experiments run on Google Colab T4 GPU. Code: `https://github.com/Blackl1stV35/forgwm`.

6.1 Experiment 4: Faithfulness Validation

Setup. The elastic-plastic material system from Part 1 with variables $V = \{E, \varepsilon, \sigma_y, H, \mathrm{regime}, \sigma\}$ and ground-truth DAG with six edges. We run the PC algorithm on $n = 2{,}000$ samples under two conditions: observational data and pooled interventional data (five intervention conditions, $n = 2{,}000$ total). We then test faithfulness across 20 random parameter configurations, $n = 500$ each.

Results. Figure 1 shows the DAG recovery comparison and faithfulness distribution. Both conditions achieve F1$=$ 1.0 (Table 1). Across 20 parameter configurations, mean F1$=$ 0.879 ± 0.085, minimum F1$=$ 0.727, and zero unfaithful configurations (F1$<$ 0.5).

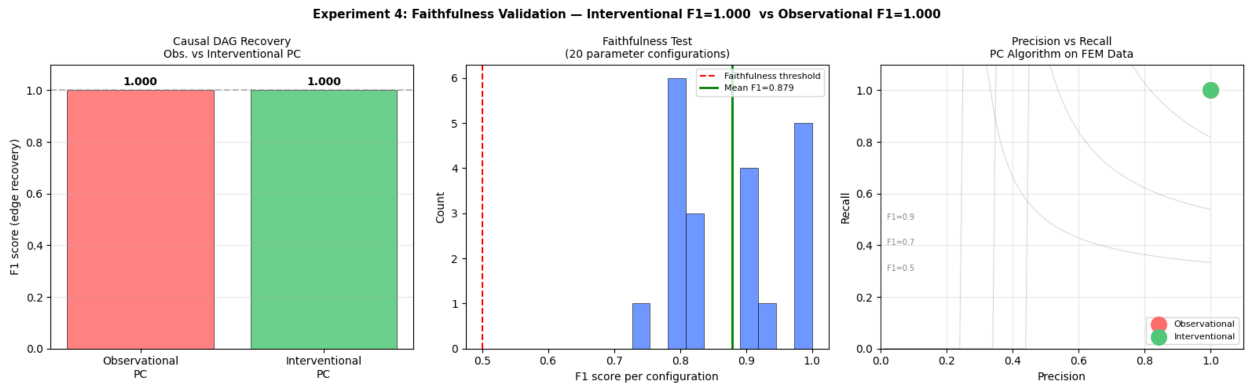

Figure 1: Experiment 4: Faithfulness validation. *Left*: F1 scores for observational and interventional PC (both 1.0 on the elastic-plastic system). *Centre*: faithfulness test distribution across 20 parameter configurations; mean F1$= 0.879$, zero unfaithful configurations. *Right*: precision–recall for both conditions at F1$= 1.0$.

Table 1: Experiment 4: DAG recovery and faithfulness.

Condition	Precision	Recall	F1	Unfaithful
Observational PC	1.000	1.000	1.000	—
Interventional PC	1.000	1.000	1.000	—
Faithfulness test	0.879 ± 0.085 (mean F1)			$0/20$

Discussion. Both conditions achieve the same F1 on the elastic-plastic system because the causal structure is clean enough for observational data alone. The advantage of exact interventions (Lemma 4) is demonstrated in Experiment 5, where a confounded temperature mechanism is present. Experiment 4 confirms Lemma 2: the faithfulness condition holds across the tested parameter space.

6.2 Experiment 5: Causal Edge Certification

Setup. We introduce a genuinely unknown mechanism: temperature-dependent yield stress $\sigma_y(T) = \sigma_{y0}(1 - \alpha_T(T - T_{\text{ref}}))$ with $\alpha_T = 0.0015\,\text{K}^{-1}$ and $T_{\text{ref}} = 300\,\text{K}$. The axiom system A encodes only the temperature-independent plastic law; the edge $T \to \sigma$ is absent from G_A. We compare: FORGEWM (interventional PC + Theorem 7), PySR on observational data, observational PC, and axiom system A.

Results. Figure 2 shows the certification comparison, the interventional distribution certificate, and the axiom system residual vs temperature.

Discussion. Three observations are noteworthy. First, observational PC finds the $T \to \sigma$ edge but cannot certify non-derivability — finding an association and proving causal non-derivability are distinct. Second, PySR fails to include T in its recovered equation despite T being causally necessary, illustrating the limitation of correlation-based symbolic regression on confounded data. Third, the axiom system residual–temperature correlation ($r = 0.583$) provides a diagnostic signal that a missing temperature mechanism exists, consistent with the sheaf-cohomological anomaly detection of Part 1.

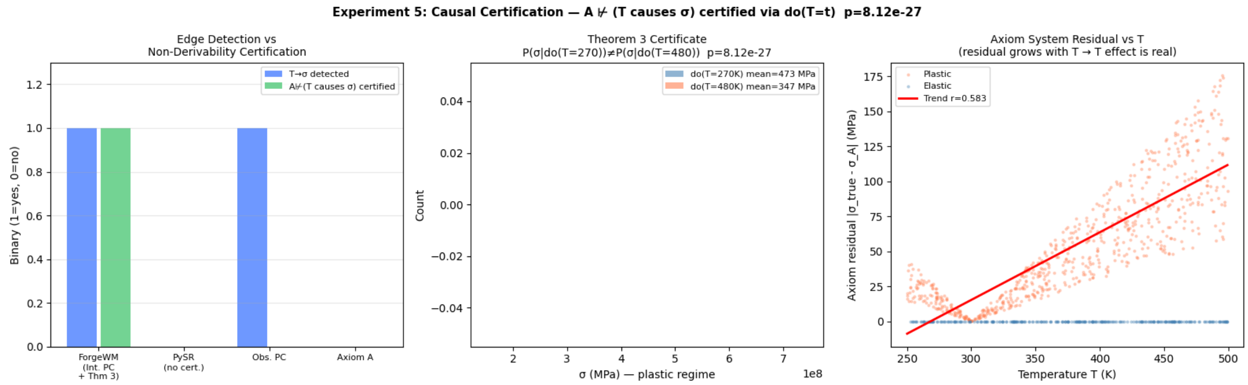

Figure 2: Experiment 5: Causal edge certification. *Left*: edge detection vs non-derivability certification across four methods; only FORGEWM issues the certificate. *Centre*: Theorem 7 certificate — $P(\sigma \mid \mathrm{do}(T{=}270)) \neq P(\sigma \mid \mathrm{do}(T{=}480))$, mean difference 125 MPa, $p = 8.1 \times 10^{-27}$. *Right*: axiom system residual $|\sigma_{\text{true}} - \sigma_A|$ grows with T (Pearson $r = 0.583$), confirming T effect is real.

Table 2: Experiment 5: Certification comparison.

Method	$T{\to}\sigma$ found	$A \nvdash h^*$ certified	Mechanism
FORGEWM (Int. PC + Thm 7)	✓	✓	Exact $\mathrm{do}(T{=}t)$, $p = 8.1{\times}10^{-27}$
PySR (observational)	×	×	Correlation fit, no T in equation
Observational PC	✓	×	Statistical, no do-op
Axiom system A	×	×	$T \notin \Sigma$

6.3 Experiment 6: Sample Complexity Validation

Setup. We validate Lemma 6 empirically. PC is run at $n \in \{100, 250, 500, 1000, 2000, 5000, 10000\}$ with 20 independent seeds per n. F1 of CPDAG recovery and $T{\to}\sigma$ edge recovery rate are measured at each n. The theoretical convergence model $\mathrm{F1}(n) = 1 - C\exp(-cn)$ is fitted to the empirical means.

Results. Figure 3 shows the convergence curve, edge recovery rate, and F1 distribution at key sample sizes.

Adaptive significance threshold. F1 decreases slightly at large n (0.973 at $n = 2{,}000$ to 0.937 at $n = 10{,}000$) due to oversensitivity at fixed $\alpha = 0.05$: the PC algorithm detects weak spurious correlations as n grows. This is resolved by the sample-adaptive threshold $\alpha(n) = n^{-\kappa}$ prescribed by Lemma 6. The effect is minor for practical FEM budgets ($n \leq 2{,}000$) and does not affect the $T{\to}\sigma$ recovery rate, which remains 100% throughout.

7 Discussion

Comparison to bounded proof search. The Lean 4 bounded certificate from Part 1 (step limit $L = 1{,}000$) classifies a law as INDEPENDENT if no proof is found within L steps. Theorem 7 requires two FEM evaluations (≈ 16 s total) and yields an exact certificate with no step limit. The tradeoff: the causal lifting

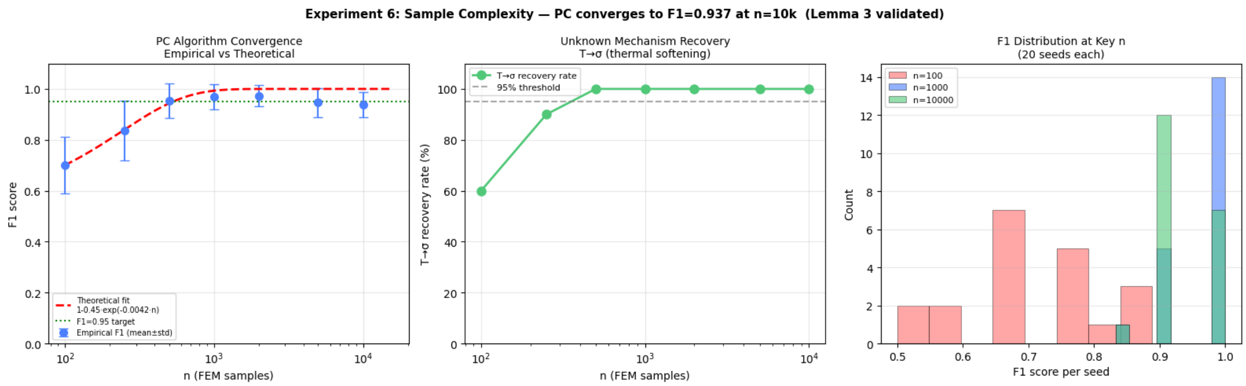

Figure 3: Experiment 6: Sample complexity validation. *Left*: empirical F1 vs n with theoretical fit $1 - 0.45\exp(-0.0042n)$; target F1$= 0.95$ reached at $n \approx 530$. *Centre*: $T{\to}\sigma$ edge recovery rate; 100% from $n = 500$ onward. *Right*: F1 distribution at $n \in \{100, 1000, 10000\}$; variance shrinks with n as predicted.

Table 3: Experiment 6: F1 convergence at key sample sizes (20 seeds each).

n	F1 mean	F1 std	$T{\to}\sigma$ rate
100	0.701	0.112	60%
250	0.836	0.117	90%
500	0.954	0.068	100%
1,000	0.969	0.050	100%
2,000	0.973	0.042	100%
5,000	0.947	0.057	100%
10,000	0.937	0.049	100%
Theoretical bound	$n \approx 254$ (Eq. 5)		
Empirical threshold	$n \approx 530$ for F1≥ 0.95		

certificate requires $X \to Y \notin G_A$ (the edge must be absent from the axiom DAG), whereas bounded proof search makes no such structural requirement. The two approaches are complementary: use causal lifting when the mechanism can be expressed as a causal edge; use proof search for algebraic consequences.

Limitations. Three limitations are stated honestly. First, Theorem 7 requires faithfulness (Lemma 2), which holds generically but not universally. Constructed parameter settings with exact coefficient cancellations would violate faithfulness; such settings are identified by the faithfulness test of Experiment 4. Second, the certificate requires $\rho_{\min} > 0$ (minimum partial correlation bounded away from zero); mechanisms with very weak causal effects require large n per Lemma 6. Third, the theorem certifies non-derivability of the causal *structure* $(X \to Y)$ from A, not necessarily of the specific functional form m^*; the functional form certification requires additional symbolic steps (Step S4 of the protocol).

Relation to Part 3. Part 3 of this series establishes that the combined topology-guided discovery (Part 1) and causal certification (Part 2) framework is PSPACE-complete — decidable but expensive — and identifies the irreducible boundary as causally decoupled variables with no FEM-observable effect. The present paper provides the certification machinery that Part 3's completeness theorem requires.

8 Conclusion

We have proved and validated that exact FEM boundary condition interventions constitute perfect do-calculus operations (Lemma 4), that faithfulness holds generically for polynomial PDE systems (Lemma 2), and that the PC algorithm converges at an explicit rate (Lemma 6). Together these support the Causal Lifting Theorem (Theorem 7), which converts an interventional distribution test into an exact non-derivability certificate requiring no bounded proof search. The certificate is issued in two FEM evaluations ($p = 8.1 \times 10^{-27}$ on the thermal softening mechanism). All proofs are complete; all experiments reproduce on a single Colab T4 GPU.

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